Chapter 4

Integers and rationals

4.1 The integers

In Chapter 2 we built up most of the basic properties of the natural number system, but we have reached the limits of what one can do with just addition and multiplication. We would now like to introduce a new operation, that of subtraction, but to do that properly we will have to pass from the natural number system to a larger number system, that of the *integers*.

Informally, the integers are what you can get by subtracting two natural numbers; for instance, $3-5$ should be an integer, as should $6-2$. This is not a complete definition of the integers, because (a) it doesn't say when two differences are equal (for instance we should know why $3-5$ is equal to $2-4$, but is not equal to $1-6$), and (b) it doesn't say how to do arithmetic on these differences (how does one add $3-5$ to $6-2$?). Furthermore, (c) this definition is circular because it requires a notion of subtraction, which we can only adequately define once the integers are constructed. Fortunately, because of our prior experience with integers we know what the answers to these questions should be. To answer (a), we know from our advanced knowledge in algebra that $a-b=c-d$ happens exactly when $a+d=c+b$, so we can characterize equality of differences using only the concept of addition. Similarly, to answer (b) we know from algebra that $(a-b)+(c-d)=(a+c)-(b+d)$ and that $(a-b)(c-d)=(ac+bd)-(ad+bc)$. So we will take advantage of our foreknowledge by building all this into the *definition* of the integers, as we shall do shortly.

We still have to resolve (c). To get around this problem we will use the following work-around: we will temporarily write integers not as a difference $a-b$, but instead use a new notation a—b to define integers,

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where the — is a meaningless place-holder, similar to the comma in the Cartesian co-ordinate notation (x, y) for points in the plane. Later when we define subtraction we will see that $a—b$ is in fact equal to $a - b$, and so we can discard the notation —; it is only needed right now to avoid circularity. (These devices are similar to the scaffolding used to construct a building; they are temporarily essential to make sure the building is built correctly, but once the building is completed they are thrown away and never used again.) This may seem unnecessarily complicated in order to define something that we already are very familiar with, but we will use this device again to construct the rationals, and knowing these kinds of constructions will be very helpful in later chapters.

Definition 4.1.1 (Integers)**.** An *integer* is an expression[1] of the form $a—b$, where a and b are natural numbers. Two integers are considered to be equal, $a—b = c—d$, if and only if $a + d = c + b$. We let $\mathbf{Z}$ denote the set of all integers.

Thus for instance $3—5$ is an integer, and is equal to $2—4$, because $3 + 4 = 2 + 5$. On the other hand, $3—5$ is not equal to $2—3$ because $3+3 \neq 2+5$. This notation is strange looking, and has a few deficiencies; for instance, 3 is not yet an integer, because it is not of the form $a—b$! We will rectify these problems later.

We have to check that this is a legitimate notion of equality. We need to verify the reflexivity, symmetry, transitivity, and substitution axioms (see Section A.7). We leave reflexivity and symmetry to Exercise 4.1.1 and instead verify the transitivity axiom. Suppose we know that $a—b = c—d$ and $c—d = e—f$. Then we have $a + d = c + b$ and $c+f = d+e$. Adding the two equations together we obtain $a+d+c+f = c+b+d+e$. By Proposition 2.2.6 we can cancel the c and d, obtaining $a + f = b + e$, i.e., $a—b = e—f$. Thus the cancellation law was needed to make sure that our notion of equality is sound. As for the substitution axiom, we cannot verify it at this stage because we have not yet defined any operations on the integers. However, when we do define

[1] In the language of set theory, what we are doing here is starting with the space $\mathbf{N} \times \mathbf{N}$ of ordered pairs (a, b) of natural numbers. Then we place an *equivalence relation* $\sim$ on these pairs by declaring $(a, b) \sim (c, d)$ iff $a+d = c+b$. The set-theoretic interpretation of the symbol $a—b$ is that it is the space of all pairs equivalent to (a, b): $a—b := \{(c, d) \in \mathbf{N} \times \mathbf{N} : (a, b) \sim (c, d)\}$. However, this interpretation plays no rôle in how we manipulate the integers and we will not refer to it again. A similar set-theoretic interpretation can be given to the construction of the rational numbers later in this chapter, or the real numbers in the next chapter.

our basic operations on the integers, such as addition, multiplication, and order, we will have to verify the substitution axiom at that time in order to ensure that the definition is valid. (We will only need to do this for the basic operations; more advanced operations on the integers, such as exponentiation, will be defined in terms of the basic ones, and so we do not need to re-verify the substitution axiom for the advanced operations.)

Now we define two basic arithmetic operations on integers: addition and multiplication.

Definition 4.1.2. The sum of two integers, $(a—b)+(c—d)$, is defined by the formula

$$(a—b) + (c—d) := (a + c)—(b + d).$$

The product of two integers, $(a—b) \times (c—d)$, is defined by

$$(a—b) \times (c—d) := (ac + bd)—(ad + bc).$$

Thus for instance, $(3—5) + (1—4)$ is equal to $(4—9)$. There is however one thing we have to check before we can accept these definitions - we have to check that if we replace one of the integers by an equal integer, that the sum or product does not change. For instance, $(3—5)$ is equal to $(2—4)$, so $(3—5)+(1—4)$ ought to have the same value as $(2—4) + (1—4)$, otherwise this would not give a consistent definition of addition. Fortunately, this is the case:

Lemma 4.1.3 (Addition and multiplication are well-defined)**.** *Let a, b, a', b', c, d be natural numbers. If $(a—b) = (a'—b')$, then $(a—b) + (c—d) = (a'—b') + (c—d)$ and $(a—b) \times (c—d) = (a'—b') \times (c—d)$, and also $(c—d) + (a—b) = (c—d) + (a'—b')$ and $(c—d) \times (a—b) = (c—d) \times (a'—b')$. Thus addition and multiplication are well-defined operations (equal inputs give equal outputs).*

Proof. To prove that $(a—b)+(c—d) = (a'—b')+(c—d)$, we evaluate both sides as $(a + c)—(b + d)$ and $(a' + c)—(b' + d)$. Thus we need to show that $a+c+b'+d = a'+c+b+d$. But since $(a—b) = (a'—b')$, we have $a + b' = a' + b$, and so by adding $c + d$ to both sides we obtain the claim. Now we show that $(a—b) \times (c—d) = (a'—b') \times (c—d)$. Both sides evaluate to $(ac + bd)—(ad + bc)$ and $(a'c + b'd)—(a'd + b'c)$, so we have to show that $ac + bd + a'd + b'c = a'c + b'd + ad + bc$. But the left-hand side factors as $c(a + b') + d(a' + b)$, while the right factors as

$c(a' + b) + d(a + b')$. Since $a + b' = a' + b$, the two sides are equal. The other two identities are proven similarly. □

The integers n—0 behave in the same way as the natural numbers n; indeed one can check that $(n—0) + (m—0) = (n + m)—0$ and $(n—0) \times (m—0) = nm—0$. Furthermore, $(n—0)$ is equal to $(m—0)$ if and only if $n = m$. (The mathematical term for this is that there is an *isomorphism* between the natural numbers n and those integers of the form n—0.) Thus we may *identify* the natural numbers with integers by setting $n \equiv n—0$; this does not affect our definitions of addition or multiplication or equality since they are consistent with each other. For instance the natural number 3 is now considered to be the same as the integer 3—0, thus $3 = 3—0$. In particular 0 is equal to 0—0 and 1 is equal to 1—0. Of course, if we set n equal to n—0, then it will also be equal to any other integer which is equal to n—0, for instance 3 is equal not only to 3—0, but also to 4—1, 5—2, etc.

We can now define incrementation on the integers by defining $x{+}{+} := x + 1$ for any integer x; this is of course consistent with our definition of the increment operation for natural numbers. However, this is no longer an important operation for us, as it has been now superceded by the more general notion of addition.

Now we consider some other basic operations on the integers.

Definition 4.1.4 (Negation of integers)**.** If $(a—b)$ is an integer, we define the negation $-(a—b)$ to be the integer $(b—a)$. In particular if $n = n—0$ is a positive natural number, we can define its negation $-n = 0—n$.

For instance $-(3—5) = (5—3)$. One can check this definition is well-defined (Exercise 4.1.2).

We can now show that the integers correspond exactly to what we expect.

Lemma 4.1.5 (Trichotomy of integers)**.** *Let x be an integer. Then exactly one of the following three statements is true: (a) x is zero; (b) x is equal to a positive natural number n; or (c) x is the negation $-n$ of a positive natural number n.*

Proof. We first show that at least one of (a), (b), (c) is true. By definition, $x = a—b$ for some natural numbers a, b. We have three cases: $a > b$, $a = b$, or $a < b$. If $a > b$ then $a = b + c$ for some positive natural number c, which means that $a—b = c—0 = c$, which is (b). If $a = b$,

then $a—b = a—a = 0—0 = 0$, which is (a). If $a < b$, then $b > a$, so that $b—a = n$ for some natural number n by the previous reasoning, and thus $a—b = -n$, which is (c).

Now we show that no more than one of (a), (b), (c) can hold at a time. By definition, a positive natural number is non-zero, so (a) and (b) cannot simultaneously be true. If (a) and (c) were simultaneously true, then $0 = -n$ for some positive natural n; thus $(0—0) = (0—n)$, so that $0 + n = 0 + 0$, so that $n = 0$, a contradiction. If (b) and (c) were simultaneously true, then $n = -m$ for some positive n, m, so that $(n—0) = (0—m)$, so that $n+m = 0+0$, which contradicts Proposition 2.2.8. Thus exactly one of (a), (b), (c) is true for any integer x. □

If n is a positive natural number, we call $-n$ a *negative integer*. Thus every integer is positive, zero, or negative, but not more than one of these at a time.

One could well ask why we don't use Lemma 4.1.5 to *define* the integers; i.e., why didn't we just say an integer is anything which is either a positive natural number, zero, or the negative of a natural number. The reason is that if we did so, the rules for adding and multiplying integers would split into many different cases (e.g., negative times positive equals positive; negative plus positive is either negative, positive, or zero, depending on which term is larger, etc.) and to verify all the properties would end up being much messier.

We now summarize the algebraic properties of the integers.

Proposition 4.1.6 (Laws of algebra for integers)**.** *Let x, y, z be integers. Then we have*

$$\begin{aligned} x + y &= y + x \\ (x + y) + z &= x + (y + z) \\ x + 0 = 0 + x &= x \\ x + (-x) = (-x) + x &= 0 \\ xy &= yx \\ (xy)z &= x(yz) \\ x1 = 1x &= x \\ x(y + z) &= xy + xz \\ (y + z)x &= yx + zx. \end{aligned}$$

Remark 4.1.7. The above set of nine identities have a name; they are asserting that the integers form a *commutative ring*. (If one deleted the

identity $xy = yx$, then they would only assert that the integers form a *ring*). Note that some of these identities were already proven for the natural numbers, but this does not automatically mean that they also hold for the integers because the integers are a larger set than the natural numbers. On the other hand, this proposition supercedes many of the propositions derived earlier for natural numbers.

Proof. There are two ways to prove these identities. One is to use Lemma 4.1.5 and split into a lot of cases depending on whether x, y, z are zero, positive, or negative. This becomes very messy. A shorter way is to write $x = (a—b)$, $y = (c—d)$, and $z = (e—f)$ for some natural numbers a, b, c, d, e, f, and expand these identities in terms of a, b, c, d, e, f and use the algebra of the natural numbers. This allows each identity to be proven in a few lines. We shall just prove the longest one, namely $(xy)z = x(yz)$:

$$\begin{aligned}
(xy)z &= ((a—b)(c—d))\,(e—f) \\
&= ((ac+bd)—(ad+bc))\,(e—f) \\
&= ((ace+bde+adf+bcf)—(acf+bdf+ade+bce))\,; \\
x(yz) &= (a—b)\,((c—d)(e—f)) \\
&= (a—b)\,((ce+df)—(cf+de)) \\
&= ((ace+adf+bcf+bde)—(acf+ade+bce+bdf))
\end{aligned}$$

and so one can see that $(xy)z$ and $x(yz)$ are equal. The other identities are proven in a similar fashion; see Exercise 4.1.4. □

We now define the operation of *subtraction* $x - y$ of two integers by the formula

$$x - y := x + (-y).$$

We do not need to verify the substitution axiom for this operation, since we have defined subtraction in terms of two other operations on integers, namely addition and negation, and we have already verified that those operations are well-defined.

One can easily check now that if a and b are natural numbers, then

$$a - b = a + -b = (a—0) + (0—b) = a—b,$$

and so $a—b$ is just the same thing as $a - b$. Because of this we can now discard the — notation, and use the familiar operation of subtraction

instead. (As remarked before, we could not use subtraction immediately because it would be circular.)

We can now generalize Lemma 2.3.3 and Corollary 2.3.7 from the natural numbers to the integers:

Proposition 4.1.8 (Integers have no zero divisors)**.** *Let a and b be integers such that $ab = 0$. Then either $a = 0$ or $b = 0$ (or both).*

Proof. See Exercise 4.1.5. □

Corollary 4.1.9 (Cancellation law for integers)**.** *If a, b, c are integers such that $ac = bc$ and c is non-zero, then $a = b$.*

Proof. See Exercise 4.1.6. □

We now extend the notion of order, which was defined on the natural numbers, to the integers by repeating the definition verbatim:

Definition 4.1.10 (Ordering of the integers)**.** Let n and m be integers. We say that n is *greater than or equal to* m, and write $n \geq m$ or $m \leq n$, iff we have $n = m + a$ for some natural number a. We say that n is *strictly greater than* m, and write $n > m$ or $m < n$, iff $n \geq m$ and $n \neq m$.

Thus for instance $5 > -3$, because $5 = -3 + 8$ and $5 \neq -3$. Clearly this definition is consistent with the notion of order on the natural numbers, since we are using the same definition.

Using the laws of algebra in Proposition 4.1.6 it is not hard to show the following properties of order:

Lemma 4.1.11 (Properties of order)**.** *Let a, b, c be integers.*

(*a*) *$a > b$ if and only if $a - b$ is a positive natural number.*

(*b*) (*Addition preserves order*) *If $a > b$, then $a + c > b + c$.*

(*c*) (*Positive multiplication preserves order*) *If $a > b$ and c is positive, then $ac > bc$.*

(*d*) (*Negation reverses order*) *If $a > b$, then $-a < -b$.*

(*e*) (*Order is transitive*) *If $a > b$ and $b > c$, then $a > c$.*

(*f*) (*Order trichotomy*) *Exactly one of the statements* $a > b$, $a < b$, *or* $a = b$ *is true.*

Proof. See Exercise 4.1.7. □

— Exercises —

Exercise 4.1.1. Verify that the definition of equality on the integers is both reflexive and symmetric.

Exercise 4.1.2. Show that the definition of negation on the integers is well-defined in the sense that if $(a—b) = (a'—b')$, then $-(a—b) = -(a'—b')$ (so equal integers have equal negations).

Exercise 4.1.3. Show that $(-1) \times a = -a$ for every integer a.

Exercise 4.1.4. Prove the remaining identities in Proposition 4.1.6. (Hint: one can save some work by using some identities to prove others. For instance, once you know that $xy = yx$, you get for free that $x1 = 1x$, and once you also prove $x(y + z) = xy + xz$, you automatically get $(y + z)x = yx + zx$ for free.)

Exercise 4.1.5. Prove Proposition 4.1.8. (Hint: while this proposition is not quite the same as Lemma 2.3.3, it is certainly legitimate to use Lemma 2.3.3 in the course of proving Proposition 4.1.8.)

Exercise 4.1.6. Prove Corollary 4.1.9. (Hint: there are two ways to do this. One is to use Proposition 4.1.8 to conclude that $a - b$ must be zero. Another way is to combine Corollary 2.3.7 with Lemma 4.1.5.)

Exercise 4.1.7. Prove Lemma 4.1.11. (Hint: use the first part of this lemma to prove all the others.)

Exercise 4.1.8. Show that the principle of induction (Axiom 2.5) does not apply directly to the integers. More precisely, give an example of a property $P(n)$ pertaining to an integer n such that $P(0)$ is true, and that $P(n)$ implies $P(n{+}{+})$ for all integers n, but that $P(n)$ is not true for all integers n. Thus induction is not as useful a tool for dealing with the integers as it is with the natural numbers. (The situation becomes even worse with the rational and real numbers, which we shall define shortly.)

4.2 The rationals

We have now constructed the integers, with the operations of addition, subtraction, multiplication, and order and verified all the expected algebraic and order-theoretic properties. Now we will use a similar construction to build the rationals, adding division to our mix of operations.

Just like the integers were constructed by subtracting two natural numbers, the rationals can be constructed by dividing two integers,

though of course we have to make the usual caveat that the denominator should be non-zero[2]. Of course, just as two differences $a - b$ and $c - d$ can be equal if $a + d = c + b$, we know (from more advanced knowledge) that two quotients a/b and c/d can be equal if $ad = bc$. Thus, in analogy with the integers, we create a new meaningless symbol $//$ (which will eventually be superceded by division), and define

Definition 4.2.1. A *rational number* is an expression of the form $a//b$, where a and b are integers and b is non-zero; $a//0$ is not considered to be a rational number. Two rational numbers are considered to be equal, $a//b = c//d$, if and only if $ad = cb$. The set of all rational numbers is denoted $\mathbf{Q}$.

Thus for instance $3//4 = 6//8 = -3//-4$, but $3//4 \neq 4//3$. This is a valid definition of equality (Exercise 4.2.1). Now we need a notion of addition, multiplication, and negation. Again, we will take advantage of our pre-existing knowledge, which tells us that $a/b + c/d$ should equal $(ad+bc)/(bd)$ and that $a/b * c/d$ should equal ac/bd, while $-(a/b)$ equals $(-a)/b$. Motivated by this foreknowledge, we define

Definition 4.2.2. If $a//b$ and $c//d$ are rational numbers, we define their sum

$$(a//b) + (c//d) := (ad + bc)//(bd)$$

their product

$$(a//b) * (c//d) := (ac)//(bd)$$

and the negation

$$-(a//b) := (-a)//b.$$

Note that if b and d are non-zero, then bd is also non-zero, by Proposition 4.1.8, so the sum or product of two rational numbers remains a rational number.

Lemma 4.2.3. *The sum, product, and negation operations on rational numbers are well-defined, in the sense that if one replaces $a//b$ with another rational number $a'//b'$ which is equal to $a//b$, then the output of the above operations remains unchanged, and similarly for $c//d$.*

[2]There is no reasonable way we can divide by zero, since one cannot have both the identities $(a/b) * b = a$ and $c * 0 = 0$ hold simultaneously if b is allowed to be zero. However, we can eventually get a reasonable notion of dividing by a quantity which *approaches* zero - think of L'Hôpital's rule (see Section 10.5), which suffices for doing things like defining differentiation.

Proof. We just verify this for addition; we leave the remaining claims to Exercise 4.2.2. Suppose $a//b = a'//b'$, so that b and b' are non-zero and $ab' = a'b$. We now show that $a//b + c//d = a'//b' + c//d$. By definition, the left-hand side is $(ad + bc)//bd$ and the right-hand side is $(a'd + b'c)//b'd$, so we have to show that

$$(ad + bc)b'd = (a'd + b'c)bd,$$

which expands to

$$ab'd^2 + bb'cd = a'bd^2 + bb'cd.$$

But since $ab' = a'b$, the claim follows. Similarly if one replaces $c//d$ by $c'//d'$. □

We note that the rational numbers $a//1$ behave in a manner identical to the integers a:

$$\begin{aligned}(a//1) + (b//1) &= (a + b)//1;\\ (a//1) \times (b//1) &= (ab//1);\\ -(a//1) &= (-a)//1.\end{aligned}$$

Also, $a//1$ and $b//1$ are only equal when a and b are equal. Because of this, we will identify a with $a//1$ for each integer a: $a \equiv a//1$; the above identities then guarantee that the arithmetic of the integers is consistent with the arithmetic of the rationals. Thus just as we embedded the natural numbers inside the integers, we embed the integers inside the rational numbers. In particular, all natural numbers are rational numbers, for instance 0 is equal to $0//1$ and 1 is equal to $1//1$.

Observe that a rational number $a//b$ is equal to $0 = 0//1$ if and only if $a \times 1 = b \times 0$, i.e., if the numerator a is equal to 0. Thus if a and b are non-zero then so is $a//b$.

We now define a new operation on the rationals: reciprocal. If $x = a//b$ is a non-zero rational (so that $a, b \neq 0$) then we define the *reciprocal* x^{-1} of x to be the rational number $x^{-1} := b//a$. It is easy to check that this operation is consistent with our notion of equality: if two rational numbers $a//b$, $a'//b'$ are equal, then their reciprocals are also equal. (In contrast, an operation such as "numerator" is not well-defined: the rationals $3//4$ and $6//8$ are equal, but have unequal numerators, so we have to be careful when referring to such terms as "the numerator of x".) We however leave the reciprocal of 0 undefined.

We now summarize the algebraic properties of the rationals.

Proposition 4.2.4 (Laws of algebra for rationals)**.** *Let x, y, z be rationals. Then the following laws of algebra hold:*

$$\begin{aligned} x + y &= y + x \\ (x + y) + z &= x + (y + z) \\ x + 0 = 0 + x &= x \\ x + (-x) = (-x) + x &= 0 \\ xy &= yx \\ (xy)z &= x(yz) \\ x1 = 1x &= x \\ x(y + z) &= xy + xz \\ (y + z)x &= yx + zx. \end{aligned}$$

If x is non-zero, we also have

$$xx^{-1} = x^{-1}x = 1.$$

Remark 4.2.5. The above set of ten identities have a name; they are asserting that the rationals $\mathbf{Q}$ form a *field*. This is better than being a commutative ring because of the tenth identity $xx^{-1} = x^{-1}x = 1$. Note that this proposition supercedes Proposition 4.1.6.

Proof. To prove this identity, one writes $x = a//b$, $y = c//d$, $z = e//f$ for some integers a, c, e and non-zero integers b, d, f, and verifies each identity in turn using the algebra of the integers. We shall just prove the longest one, namely $(x + y) + z = x + (y + z)$:

$$\begin{aligned} (x + y) + z &= ((a//b) + (c//d)) + (e//f) \\ &= ((ad + bc)//bd) + (e//f) \\ &= (adf + bcf + bde)//bdf; \\ x + (y + z) &= (a//b) + ((c//d) + (e//f)) \\ &= (a//b) + ((cf + de)//df) \\ &= (adf + bcf + bde)//bdf \end{aligned}$$

and so one can see that $(x + y) + z$ and $x + (y + z)$ are equal. The other identities are proven in a similar fashion and are left to Exercise 4.2.3. $\square$

We can now define the *quotient* x/y of two rational numbers x and y, *provided that* y is non-zero, by the formula

$$x/y := x \times y^{-1}.$$

Thus, for instance

$$(3//4)/(5//6) = (3//4) \times (6//5) = (18//20) = (9//10).$$

Using this formula, it is easy to see that $a/b = a//b$ for every integer a and every non-zero integer b. Thus we can now discard the // notation, and use the more customary a/b instead of $a//b$.

In a similar spirit, we define subtraction on the rationals by the formula

$$x - y := x + (-y),$$

just as we did with the integers.

Proposition 4.2.4 allows us to use all the normal rules of algebra; we will now proceed to do so without further comment.

In the previous section we organized the integers into positive, zero, and negative numbers. We now do the same for the rationals.

Definition 4.2.6. A rational number x is said to be *positive* iff we have $x = a/b$ for some positive integers a and b. It is said to be *negative* iff we have $x = -y$ for some positive rational y (i.e., $x = (-a)/b$ for some positive integers a and b).

Thus for instance, every positive integer is a positive rational number, and every negative integer is a negative rational number, so our new definition is consistent with our old one.

Lemma 4.2.7 (Trichotomy of rationals)**.** *Let x be a rational number. Then exactly one of the following three statements is true: (a) x is equal to 0. (b) x is a positive rational number. (c) x is a negative rational number.*

Proof. See Exercise 4.2.4. □

Definition 4.2.8 (Ordering of the rationals)**.** Let x and y be rational numbers. We say that $x > y$ iff $x - y$ is a positive rational number, and $x < y$ iff $x - y$ is a negative rational number. We write $x \geq y$ iff either $x > y$ or $x = y$, and similarly define $x \leq y$.

Proposition 4.2.9 (Basic properties of order on the rationals)**.** *Let x, y, z be rational numbers. Then the following properties hold.*

(a) *(Order trichotomy) Exactly one of the three statements $x = y$, $x < y$, or $x > y$ is true.*

(b) *(Order is anti-symmetric) One has $x < y$ if and only if $y > x$.*

(c) *(Order is transitive) If $x < y$ and $y < z$, then $x < z$.*

(d) *(Addition preserves order) If $x < y$, then $x + z < y + z$.*

(e) *(Positive multiplication preserves order) If $x < y$ and z is positive, then $xz < yz$.*

Proof. See Exercise 4.2.5. □

Remark 4.2.10. The above five properties in Proposition 4.2.9, combined with the field axioms in Proposition 4.2.4, have a name: they assert that the rationals $\mathbf{Q}$ form an *ordered field*. It is important to keep in mind that Proposition 4.2.9(e) only works when z is positive, see Exercise 4.2.6.

— Exercises —

Exercise 4.2.1. Show that the definition of equality for the rational numbers is reflexive, symmetric, and transitive. (Hint: for transitivity, use Corollary 4.1.9.)

Exercise 4.2.2. Prove the remaining components of Lemma 4.2.3.

Exercise 4.2.3. Prove the remaining components of Proposition 4.2.4. (Hint: as with Proposition 4.1.6, you can save some work by using some identities to prove others.)

Exercise 4.2.4. Prove Lemma 4.2.7. (Note that, as in Proposition 2.2.13, you have to prove two different things: firstly, that *at least* one of (a), (b), (c) is true; and secondly, that *at most* one of (a), (b), (c) is true.)

Exercise 4.2.5. Prove Proposition 4.2.9.

Exercise 4.2.6. Show that if x, y, z are rational numbers such that $x < y$ and z is *negative*, then $xz > yz$.

4.3 Absolute value and exponentiation

We have already introduced the four basic arithmetic operations of addition, subtraction, multiplication, and division on the rationals. (Recall that subtraction and division came from the more primitive notions of negation and reciprocal by the formulae $x - y := x + (-y)$ and

$x/y := x \times y^{-1}$.) We also have a notion of order $<$, and have organized the rationals into the positive rationals, the negative rationals, and zero. In short, we have shown that the rationals $\mathbf{Q}$ form an *ordered field*.

One can now use these basic operations to construct more operations. There are many such operations we can construct, but we shall just introduce two particularly useful ones: absolute value and exponentiation.

Definition 4.3.1 (Absolute value)**.** If x is a rational number, the *absolute value* $|x|$ of x is defined as follows. If x is positive, then $|x| := x$. If x is negative, then $|x| := -x$. If x is zero, then $|x| := 0$.

Definition 4.3.2 (Distance)**.** Let x and y be rational numbers. The quantity $|x - y|$ is called the *distance between x and y* and is sometimes denoted $d(x, y)$, thus $d(x, y) := |x - y|$. For instance, $d(3, 5) = 2$.

Proposition 4.3.3 (Basic properties of absolute value and distance)**.** *Let x, y, z be rational numbers.*

(*a*) (*Non-degeneracy of absolute value*) *We have $|x| \geq 0$. Also, $|x| = 0$ if and only if x is 0.*

(*b*) (*Triangle inequality for absolute value*) *We have $|x+y| \leq |x|+|y|$.*

(*c*) *We have the inequalities $-y \leq x \leq y$ if and only if $y \geq |x|$. In particular, we have $-|x| \leq x \leq |x|$.*

(*d*) (*Multiplicativity of absolute value*) *We have $|xy| = |x|\ |y|$. In particular, $|-x| = |x|$.*

(*e*) (*Non-degeneracy of distance*) *We have $d(x, y) \geq 0$. Also, $d(x, y) = 0$ if and only if $x = y$.*

(*f*) (*Symmetry of distance*) $d(x, y) = d(y, x)$.

(*g*) (*Triangle inequality for distance*) $d(x, z) \leq d(x, y) + d(y, z)$.

Proof. See Exercise 4.3.1. □

Absolute value is useful for measuring how "close" two numbers are. Let us make a somewhat artificial definition:

Definition 4.3.4 (ε-closeness)**.** Let $\varepsilon > 0$ be a rational number, and let x, y be rational numbers. We say that y is *ε-close* to x iff we have $d(y, x) \leq \varepsilon$.

Remark 4.3.5. This definition is not standard in mathematics textbooks; we will use it as "scaffolding" to construct the more important notions of limits (and of Cauchy sequences) later on, and once we have those more advanced notions we will discard the notion of ε-close.

Examples 4.3.6. The numbers 0.99 and 1.01 are 0.1-close, but they are not 0.01 close, because $d(0.99, 1.01) = |0.99 - 1.01| = 0.02$ is larger than 0.01. The numbers 2 and 2 are ε-close for every positive ε.

We do not bother defining a notion of ε-close when ε is zero or negative, because if ε is zero then x and y are only ε-close when they are equal, and when ε is negative then x and y are never ε-close. (In any event it is a long-standing tradition in analysis that the Greek letters ε, δ should only denote small positive numbers.)

Some basic properties of ε-closeness are the following.

Proposition 4.3.7. *Let x, y, z, w be rational numbers.*

(a) *If $x = y$, then x is ε-close to y for every $\varepsilon > 0$. Conversely, if x is ε-close to y for every $\varepsilon > 0$, then we have $x = y$.*

(b) *Let $\varepsilon > 0$. If x is ε-close to y, then y is ε-close to x.*

(c) *Let $\varepsilon, \delta > 0$. If x is ε-close to y, and y is δ-close to z, then x and z are $(\varepsilon + \delta)$-close.*

(d) *Let $\varepsilon, \delta > 0$. If x and y are ε-close, and z and w are δ-close, then $x + z$ and $y + w$ are $(\varepsilon + \delta)$-close, and $x - z$ and $y - w$ are also $(\varepsilon + \delta)$-close.*

(e) *Let $\varepsilon > 0$. If x and y are ε-close, they are also ε'-close for every $\varepsilon' > \varepsilon$.*

(f) *Let $\varepsilon > 0$. If y and z are both ε-close to x, and w is between y and z (i.e., $y \leq w \leq z$ or $z \leq w \leq y$), then w is also ε-close to x.*

(g) *Let $\varepsilon > 0$. If x and y are ε-close, and z is non-zero, then xz and yz are $\varepsilon|z|$-close.*

(h) *Let $\varepsilon, \delta > 0$. If x and y are ε-close, and z and w are δ-close, then xz and yw are $(\varepsilon|z| + \delta|x| + \varepsilon\delta)$-close.*

Proof. We only prove the most difficult one, (h); we leave (a)-(g) to Exercise 4.3.2. Let $\varepsilon, \delta > 0$, and suppose that x and y are ε-close. If we

write $a := y - x$, then we have $y = x + a$ and that $|a| \leq \varepsilon$. Similarly, if z and w are δ-close, and we define $b := w - z$, then $w = z + b$ and $|b| \leq \delta$.

Since $y = x + a$ and $w = z + b$, we have

$$yw = (x + a)(z + b) = xz + az + xb + ab.$$

Thus

$$|yw - xz| = |az + bx + ab| \leq |az| + |bx| + |ab| = |a||z| + |b||x| + |a||b|.$$

Since $|a| \leq \varepsilon$ and $|b| \leq \delta$, we thus have

$$|yw - xz| \leq \varepsilon|z| + \delta|x| + \varepsilon\delta$$

and thus that yw and xz are $(\varepsilon|z| + \delta|x| + \varepsilon\delta)$-close. □

Remark 4.3.8. One should compare statements (a)-(c) of this proposition with the reflexive, symmetric, and transitive axioms of equality. It is often useful to think of the notion of "ε-close" as an approximate substitute for that of equality in analysis.

Now we recursively define exponentiation for natural number exponents, extending the previous definition in Definition 2.3.11.

Definition 4.3.9 (Exponentiation to a natural number)**.** Let x be a rational number. To raise x to the power 0, we define $x^0 := 1$; in particular we define $0^0 := 1$. Now suppose inductively that x^n has been defined for some natural number n, then we define $x^{n+1} := x^n \times x$.

Proposition 4.3.10 (Properties of exponentiation, I)**.** *Let x, y be rational numbers, and let n, m be natural numbers.*

(a) *We have $x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.*

(b) *Suppose $n > 0$. Then we have $x^n = 0$ if and only if $x = 0$.*

(c) *If $x \geq y \geq 0$, then $x^n \geq y^n \geq 0$. If $x > y \geq 0$ and $n > 0$, then $x^n > y^n \geq 0$.*

(d) *We have $|x^n| = |x|^n$.*

Proof. See Exercise 4.3.3. □

Now we define exponentiation for negative integer exponents.

Definition 4.3.11 (Exponentiation to a negative number)**.** Let x be a non-zero rational number. Then for any negative integer $-n$, we define $x^{-n} := 1/x^n$.

Thus for instance $x^{-3} = 1/x^3 = 1/(x \times x \times x)$. We now have x^n defined for any integer n, whether n is positive, negative, or zero. Exponentiation with integer exponents has the following properties (which supercede Proposition 4.3.10):

Proposition 4.3.12 (Properties of exponentiation, II)**.** *Let x, y be non-zero rational numbers, and let n, m be integers.*

(a) *We have $x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.*

(b) *If $x \geq y > 0$, then $x^n \geq y^n > 0$ if n is positive, and $0 < x^n \leq y^n$ if n is negative.*

(c) *If $x, y > 0$, $n \neq 0$, and $x^n = y^n$, then $x = y$.*

(d) *We have $|x^n| = |x|^n$.*

Proof. See Exercise 4.3.4. □

— Exercises —

Exercise 4.3.1. Prove Proposition 4.3.3. (Hint: while all of these claims can be proven by dividing into cases, such as when x is positive, negative, or zero, several parts of the proposition can be proven without such a tedious division into cases. For instance one can use earlier parts of the proposition to prove later ones.)

Exercise 4.3.2. Prove the remaining claims in Proposition 4.3.7.

Exercise 4.3.3. Prove Proposition 4.3.10. (Hint: use induction.)

Exercise 4.3.4. Prove Proposition 4.3.12. (Hint: induction is not suitable here. Instead, use Proposition 4.3.10.)

Exercise 4.3.5. Prove that $2^N \geq N$ for all positive integers N. (Hint: use induction.)

4.4 Gaps in the rational numbers

Imagine that we arrange the rationals on a line, arranging x to the right of y if $x > y$. (This is a non-rigorous arrangement, since we have not yet defined the concept of a line, but this discussion is only intended to motivate the more rigorous propositions below.) Inside the rationals we have the integers, which are thus also arranged on the line. Now we work out how the rationals are arranged with respect to the integers.

Proposition 4.4.1 (Interspersing of integers by rationals)**.** *Let x be a rational number. Then there exists an integer n such that $n \leq x < n+1$. In fact, this integer is unique (i.e., for each x there is only one n for which $n \leq x < n+1$). In particular, there exists a natural number N such that $N > x$ (i.e., there is no such thing as a rational number which is larger than all the natural numbers).*

Remark 4.4.2. The integer n for which $n \leq x < n+1$ is sometimes referred to as the *integer part* of x and is sometimes denoted $n = \lfloor x \rfloor$.

Proof. See Exercise 4.4.1. □

Also, between every two rational numbers there is at least one additional rational:

Proposition 4.4.3 (Interspersing of rationals by rationals)**.** *If x and y are two rationals such that $x < y$, then there exists a third rational z such that $x < z < y$.*

Proof. We set $z := (x+y)/2$. Since $x < y$, and $1/2 = 1//2$ is positive, we have from Proposition 4.2.9 that $x/2 < y/2$. If we add $y/2$ to both sides using Proposition 4.2.9 we obtain $x/2+y/2 < y/2+y/2$, i.e., $z < y$. If we instead add $x/2$ to both sides we obtain $x/2 + x/2 < y/2 + x/2$, i.e., $x < z$. Thus $x < z < y$ as desired. □

Despite the rationals having this denseness property, they are still incomplete; there are still an infinite number of "gaps" or "holes" between the rationals, although this denseness property does ensure that these holes are in some sense infinitely small. For instance, we will now show that the rational numbers do not contain any square root of two.

Proposition 4.4.4. *There does not exist any rational number x for which $x^2 = 2$.*

Proof. We only give a sketch of a proof; the gaps will be filled in Exercise 4.4.3. Suppose for sake of contradiction that we had a rational number x for which $x^2 = 2$. Clearly x is not zero. We may assume that x is positive, for if x were negative then we could just replace x by $-x$ (since $x^2 = (-x)^2$). Thus $x = p/q$ for some positive integers p, q, so $(p/q)^2 = 2$, which we can rearrange as $p^2 = 2q^2$. Define a natural number p to be *even* if $p = 2k$ for some natural number k, and *odd* if $p = 2k+1$ for some natural number k. Every natural number is either even or odd, but not both (why?). If p is odd, then p^2 is also odd (why?), which contradicts

$p^2 = 2q^2$. Thus p is even, i.e., $p = 2k$ for some natural number k. Since p is positive, k must also be positive. Inserting $p = 2k$ into $p^2 = 2q^2$ we obtain $4k^2 = 2q^2$, so that $q^2 = 2k^2$.

To summarize, we started with a pair (p, q) of positive integers such that $p^2 = 2q^2$, and ended up with a pair (q, k) of positive integers such that $q^2 = 2k^2$. Since $p^2 = 2q^2$, we have $q < p$ (why?). If we rewrite $p' := q$ and $q' := k$, we thus can pass from one solution (p, q) to the equation $p^2 = 2q^2$ to a new solution (p', q') to the same equation which has a smaller value of p. But then we can repeat this procedure again and again, obtaining a sequence (p'', q''), (p''', q'''), etc. of solutions to $p^2 = 2q^2$, each one with a smaller value of p than the previous, and each one consisting of positive integers. But this contradicts the principle of infinite descent (see Exercise 4.4.2). This contradiction shows that we could not have had a rational x for which $x^2 = 2$. □

On the other hand, we can get rational numbers which are arbitrarily close to a square root of 2:

Proposition 4.4.5. *For every rational number $\varepsilon > 0$, there exists a non-negative rational number x such that $x^2 < 2 < (x + \varepsilon)^2$.*

Proof. Let $\varepsilon > 0$ be rational. Suppose for sake of contradiction that there is no non-negative rational number x for which $x^2 < 2 < (x + \varepsilon)^2$. This means that whenever x is non-negative and $x^2 < 2$, we must also have $(x + \varepsilon)^2 < 2$ (note that $(x + \varepsilon)^2$ cannot equal 2, by Proposition 4.4.4). Since $0^2 < 2$, we thus have $\varepsilon^2 < 2$, which then implies $(2\varepsilon)^2 < 2$, and indeed a simple induction shows that $(n\varepsilon)^2 < 2$ for every natural number n. (Note that $n\varepsilon$ is non-negative for every natural number n - why?) But, by Proposition 4.4.1 we can find an integer n such that $n > 2/\varepsilon$, which implies that $n\varepsilon > 2$, which implies that $(n\varepsilon)^2 > 4 > 2$, contradicting the claim that $(n\varepsilon)^2 < 2$ for all natural numbers n. This contradiction gives the proof. □

Example 4.4.6. If[3] $\varepsilon = 0.001$, we can take $x = 1.414$, since $x^2 = 1.999396$ and $(x + \varepsilon)^2 = 2.002225$.

Proposition 4.4.5 indicates that, while the set $\mathbf{Q}$ of rationals does not actually have $\sqrt{2}$ as a member, we can get as close as we wish to

[3] We will use the decimal system for defining terminating decimals, for instance 1.414 is defined to equal the rational number 1414/1000. We defer the formal discussion on the decimal system to an Appendix (§B).

$\sqrt{2}$. For instance, the sequence of rationals

$$1.4, 1.41, 1.414, 1.4142, 1.41421, \ldots$$

seem to get closer and closer to $\sqrt{2}$, as their squares indicate:

$$1.96, 1.9881, 1.99396, 1.99996164, 1.9999899241, \ldots$$

Thus it seems that we can create a square root of 2 by taking a "limit" of a sequence of rationals. This is how we shall construct the real numbers in the next chapter. (There is another way to do so, using something called "Dedekind cuts", which we will not pursue here. One can also proceed using infinite decimal expansions, but there are some sticky issues when doing so, e.g., one has to make $0.999\ldots$ equal to $1.000\ldots$, and this approach, despite being the most familiar, is actually *more* complicated than other approaches; see the Appendix §B.)

— Exercises —

Exercise 4.4.1. Prove Proposition 4.4.1. (Hint: use Proposition 2.3.9.)

Exercise 4.4.2. A definition: a sequence $a_0, a_1, a_2, \ldots$ of numbers (natural numbers, integers, rationals, or reals) is said to be in *infinite descent* if we have $a_n > a_{n+1}$ for all natural numbers n (i.e., $a_0 > a_1 > a_2 > \ldots$).

(a) Prove the *principle of infinite descent*: that it is not possible to have a sequence of *natural numbers* which is in infinite descent. (Hint: assume for sake of contradiction that you can find a sequence of natural numbers which is in infinite descent. Since all the a_n are natural numbers, you know that $a_n \geq 0$ for all n. Now use induction to show in fact that $a_n \geq k$ for all $k \in \mathbf{N}$ and all $n \in \mathbf{N}$, and obtain a contradiction.)

(b) Does the principle of infinite descent work if the sequence a_1, a_2, $a_3, \ldots$ is allowed to take integer values instead of natural number values? What about if it is allowed to take positive rational values instead of natural numbers? Explain.

Exercise 4.4.3. Fill in the gaps marked (why?) in the proof of Proposition 4.4.4.